Abhinav Rao

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Education

University of Maryland at College Park — College Park, MD
Ph.D. student, Computer Science (Area: AI-Security, Advisor: Yizheng Chen) — August 2025 - Present

Carnegie Mellon University | Language Technologies Institute — Pittsburgh, PA
M.S. (MIIS) (AI-Ethics and Safety, Advisor: Prof. Maarten Sap) — August 2023 - December 2024

Birla Institute of Technology and Science (BITS) Pilani — Hyderabad, India
B.E. Computer Science (Graduated Early) — August 2018 - February 2022

Select Experience

CMU - Language Technologies Institute — Pittsburgh, PA
Systems Software Engineer — Jan 2025 - July 2025

- Finalist team for the first edition of Amazon NOVA AI Challenge - Trusted AI Track.
- Developed technical approach for winning proposal, securing $250,000 for the Amazon NOVA AI Challenge.
- Employed deliberative-alignment and reinforcement learning (GRPO) to enable secure code reasoning. Achieved defense success of ∼90% in final red-teaming tournament.
- Technical report here

Bell Labs — Murray Hill, NJ
Research Intern, Autonomous Systems — June 2024 - August 2024

- Constructed a code repair prototype using multi-agent pipeline with Large Language Models (LLMs).

Microsoft — Bangalore, India
Research Fellow — July 2022 - August 2023

- Worked on Responsible AI (RAI) focusing on AI Ethics and Safety. Analyzed ethical reasoning capabilities of LLMs, and their susceptibility to jailbreaks.

Microsoft Research — Bangalore, India
Research Intern — January 2022 - July 2022

- Developed a multilingual query expansion tool with embedding interpolation and topic modeling.

Nanyang Technological University, SpeechLab — Singapore
Research Intern (SpeechLab) — June 2021 - December 2021

- Extended punctuation restoration capabilities to Chinese and Malay with XLM-R. Improved F1-score by 4.2% over state-of-the-art for Chinese punctuation restoration in ASR text using a pretraining-style objective.

Publications

S=In Submission, C=Conference, W=Workshop, P=Preprint, M=Misc

[M.1] **Secure and Useful Models are Reasonable: Aligning Code Models via Utility-Preserving Reasoning** (Atharva Naik*, Alex Xie*, **Abhinav Rao***, Anmol Agarwal*, Shubham Gandhi*, Michael Hilton, Carolyn Rosé [Amazon Nova AI Proceedings 2025])

[C.1] **Tricking LLMs into Disobedience: Understanding, Analyzing, and Preventing Jailbreaks** (**Abhinav Rao**, Sachin Vashistha*, Atharva Naik*, Somak Aditya, and Monojit Choudhury [Published at LREC-CoLING 2024])

[C.2] **Ethical Reasoning over Moral Alignment: A Case and Framework for In-Context Ethical Policies in LLMs** (**Abhinav Rao***, Aditi Khandelwal*, Kumar Tanmay*, Utkarsh Agarwal*, Monojit Choudhury [Published at the Findings of EMNLP 2023, Presented as a Keynote at WiNLP])

[C.3] Normad: A benchmark for measuring the cultural adaptability of large language models (**Abhinav Rao***, Akhila Yerukola*, Vishwa Shah, Katharina Reinecke, and Maarten Sap [Published at NAACL 2025, Non-archivally @ C3NLP, ACL 2024])

[C.4] Punctuation Restoration for Singaporean Spoken Languages (**Abhinav Rao**, Thi-Nga Ho, and Eng-Siong Chng [Asia-Pacific Speech and Information Processing Association 2022])

[W.1] Less is Fed More: Sparsity Mitigates Feature Distortion in Federated Learning (Aashiq Muhamed*, Harshita Diddee*, **Abhinav Rao*** [CustomNLP4U, EMNLP 2024, Also Presented at MOOMIN, EACL 2024])

[P.3] An Emergent Mirage: Is Emergent Misalignment and Realignment Indeed a Robust Phenomenon? (**Abhinav Rao***, Liancheng Gong*, Bin Hu*, and Atharva Naik [arXiv preprint arXiv:2607.09053])

[P.2] Jailbreak Paradox: The Achilles' Heel of LLMs (**Abhinav Rao***, Monojit Choudhury*, and Somak Aditya* [arXiv preprint arXiv:2406.12702])

[J.1] MALITE: Lightweight Malware Detection and Classification for Constrained Devices (Siddharth Anand, Barsha Mitra, Soumyadeep Dey, **Abhinav Rao**, Rupsa Dhar, and Jaideep Vaidya [IEEE Transactions on Emerging Topics in Computing (TETC) 2025])

Select Research Projects

Secure and Correct Code Generation September 2025 - Present
Advisor: Prof. Yizheng Chen

- Developed and refined a 35K-prompt, multi-language reinforcement learning dataset spanning ∼30 CWEs.
- Training Qwen3-8B with RL and secure coding guidelines to improve secure code generation and secure-principle reasoning on SecCodePLT, BaxBench, and CWEval benchmark suites.

Cultural Reasoning of LLMs September 2023 - October 2024
Advisors: Prof. Maarten Sap, Prof. Katharina Reinecke

- Built a benchmark dataset of 2.6k cultural situations spanning 75 countries measuring cultural biases in LLMs
- Measured cultural adaptability of 17 language models, determining strong sycophancy and western-centric biases. [Accepted at NAACL'25] [Presented at C3NLP, ACL '24]

Jailbreaking Language Models November 2022 - November 2023
Advisors: Prof. Monojit Choudhury, Prof. Aditya Somak

- Evaluated jailbreak effectiveness against 9 different LLMs by formalizing LLM jailbreaking, showing an inverse scaling trend where GPT-3.5 is 20% more susceptible than FLAN-T5. [Lrec-CoLING'24] (Coverage: TCS Research Webinar on Genrative AI).
- Developing a theoretical framework to explain the jailbreak-paradox, explaining the inverse scaling phenomenon in toxicity/jailbreaking. (Work-in-progress covered by Analytics IndiaMag).
- Improved Bing Chat classifier performance by 5% and 17% (F1-score) for jailbreaking and content-harm detection through offline data curation.

Ethical Reasoning Capabilities of LLMs August 2022 - July 2023
Advisors: Prof. Monojit Choudhury

- Designed a framework to evaluate the ethical reasoning capabilities of Language models over increasing granularities of ethical policies. Uncovered a bias favoring western centric ethical principles in GPT-4. [EMNLP Findings '23] [Keynote at WiNLP '23]

Talks

"Amazon NOVA AI Lightning Talk: Emergent Alignment and Misalignment"
- TrustNLP, NAACL '25, Albuquerque, NM | May 2025

"NormAd: A Framework for measuring Cultural Adaptability in Language Models (co-presented with Akhila Yerukola)"
- NAACL 2025, Albuquerque, NM | May 2025

"Less is Fed More: Sparsity Mitigates Feature Distortion in Federated Learning"
- MOOMIN, EACL '24, Malta [presentation] | March 2024 (Remote)

"Punctuation Restoration for Singaporean Spoken Languages"
- APSIPA '22, Chiang-Mai, Thailand [presentation] | November 2022 (Remote)

Honours and Awards

Amazon NOVA AI Challenge - Trusted AI Grant, 2024
- Awarded $250,000 as a model developer team for the Amazon NOVA AI Challenge - Trusted AI track.

BITS Merit Scholarship, 2018, 2022
- Tuition waiver of $3300 (INR 280,000 total) awarded to the top 3%ile of students for academic excellence.

Teaching

Advanced Natural Language Processing (CMU-LTI 11711)
- Responsibilities included conducting tutorials, evaluating assignments, and helping students with the assignments and advising them on their course projects.

Academic Service

Reviewer: ACL ARR Cycles: {December 2023, December 2024, February 2026}, TPAMI 2024

Sub-Reviewer: NAACL 2022

Volunteer: Panini Linguistics Olympiad (PLO) 2023

References

Prof. Yizheng Chen - Assistant Professor, University of Maryland, College Park (yzchen@umd.edu)

Prof. Maarten Sap - Assistant Professor, Carnegie Mellon University (maartensap@cmu.edu)

Prof. Monojit Choudhury - Professor, MBZUAI, UAE (monojit.choudhury@mbzuai.ac.ae)

Prof. Somak Aditya - Assistant Professor, IIT-KGP, India (saditya@iitkgp.ac.in)